

# Ghost Serialization 1.2.2

## Complete technical manual — study and reference (A5 / mobile)

Monorepo `ghost-serializer` · version `1.2.2` · Maven `com.ghostserializer` · compile-time KSP + reflection-free runtime.

### How to read this manual

This document is meant to be studied in PART order (I → VIII) or by jumping via the index (blue links in the PDF). Tables summarize; paragraphs and JSON examples explain the why and when to use each piece. If something does not match your project, open the `.kt` file indicated under `build/generated/ksp/` after compiling.

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# **PART I — Fundamentals**

## **1. Ghost philosophy and promise**

### **What problem it solves**

When you use Gson, Moshi, or `kotlinx.serialization` with reflection or runtime metadata, every request pays a cost: finding fields, creating adapters, or walking generic trees. In APIs with thousands of requests per second, that shows up in CPU and allocations.

Ghost inverts the approach: all heavy work happens at compile time, with KSP (Kotlin Symbol Processing). You write a normal data class with one annotation; the compiler generates a `UserSerializer.kt` file with

concrete code, no reflection.

**Compile-time vs runtime (mental diagram)**

| YOUR CODE                              |   | BUILD (KSP)                       |   | RUNTIME                                                        |
|----------------------------------------|---|-----------------------------------|---|----------------------------------------------------------------|
| @GhostSerialization<br>data class User | → | UserSerializer.kt<br>(fixed code) | → | Ghost.deserialize<User>(json)<br>calls UserSerializer directly |

At runtime Ghost does not do `Class.getDeclaredFields()`, does not build a dynamic `JsonAdapter`. It only:

- 1. Looks up the serializer in cache or registry (a generated when).
- 2. Passes JSON to a byte-by-byte parser (`GhostJsonFlatReader`).
- 3. The generated serializer reads field by field and builds the object.

**Quick comparison with other libraries**

| Aspect                | Gson / Moshi (reflectio           | kotlinx.serialization | Ghost                                |
|-----------------------|-----------------------------------|-----------------------|--------------------------------------|
| ----                  | ----                              | ----                  | ----                                 |
| When code is generate | Runtime (reflect) or optional KSP | KSP                   | Always KSP                           |
| Dynamic JSON model    | Yes                               | Limited               | No (fixed models)                    |
| R8 / ProGuard         | Broad keep rules                  | Keep rules            | Direct code, little keep             |
| Typical hot path      | Adapter map                       | Encoder/Decoder       | Flat reader + monomorphic serializer |

**The performance promise (in plain language)**

- Monomorphic paths: the JIT always sees `UserSerializer.deserialize`, not a generic interface with 200 implementations. That enables inlining.
- Byte-by-byte parser: JSON lives in a `ByteArray`; no intermediate DOM tree of objects is created.
- Per-thread pools: reader and writer are reused between calls on the same thread (`ThreadLocal` on JVM).
- Registry with when: resolving `KClass` → `Serializer` is a compiled comparison chain, not a `HashMap` of strings.

**What you should NOT expect from Ghost**

- Fully dynamic JSON (`Map<String, Any>` without a type) like Jackson for admin APIs.
- Changing the schema at runtime without recompiling.
- Replacing Jackson in a Spring project where 90% of DTOs are POJOs without `@GhostSerialization`.

Ghost shines when your network models are defined and you want maximum performance on Android, JVM server, or KMP iOS.

**Complete minimal example**

```
// 1. Model (app or shared module)
@GhostSerialization
```

```
data class User(val id: Int, val name: String)

// 2. After compiling, UserSerializer.kt exists (you do not write it)

// 3. Runtime usage
val json = """"{"id":1,"name":"Ana"}""""
val user = Ghost.deserialize<User>(json)
val again = Ghost.encodeToString(user) // {"id":1,"name":"Ana"}
```

If `Ghost.deserialize` throws `NOT_FOUND`, it almost always means: missing annotation, KSP did not run, or on iOS missing `addRegistry`.

## PART II — Public API

### 2. Monorepo module map

| Module                           | Published      | Role                                 |
|----------------------------------|----------------|--------------------------------------|
| <b>ghost-api</b>                 | <b>Yes</b>     | <b>Public annotations, zero deps</b> |
| <b>ghost-compiler</b>            | <b>Yes</b>     | <b>KSP plugin (KotlinPoet)</b>       |
| <b>ghost-serialization</b>       | <b>Yes KMP</b> | <b>Runtime engine</b>                |
| <b>ghost-gradle-plugin</b>       | <b>Yes</b>     | <b>Wiring in consumer apps</b>       |
| <b>ghost-retrofit</b>            | <b>Yes JVM</b> | <b>ConverterFactory</b>              |
| <b>ghost-ktor</b>                | <b>Yes KMP</b> | <b>ContentNegotiation ghost()</b>    |
| <b>ghost-spring-boot-starter</b> | <b>Yes JVM</b> | <b>MVC + WebFlux</b>                 |
| <b>ghost-integration-test</b>    | <b>No</b>      | <b>Compiler tests</b>                |
| <b>ghost-benchmark</b>           | <b>No</b>      | <b>JVM benchmark</b>                 |
| <b>ghost-sample</b>              | <b>No</b>      | <b>Compose demo</b>                  |

Typical consumer dependency: `ghost plugin` + `ghost-serialization` (via plugin) + `ghost-compiler` (KSP).

#### What your project needs (explained)

1. Gradle plugin `com.ghostserializer.ghost` — configures KSP, adds correct dependencies for Android, JVM, or KMP.
2. `ghost-api` — annotations only (`@GhostSerialization`, etc.). Minimal weight.
3. `ghost-compiler` — KSP processor; generates `*Serializer.kt` in `build/generated/ksp/`.
4. `ghost-serialization` — runtime: `Ghost`, parsers, writers, primitives.

You do not ship the compiler in your final app as “business logic”; it only runs at compile time. The APK/AAR contains runtime + generated code.

#### Modules NOT published to Maven Central

- `ghost-integration-test`: internal regression suite (155 tests in `ciTestJvm` modules; full Linux `ciTest` is 642 — see section 23).

- ghost-benchmark: Gson/Moshi/Kotlinx vs Ghost comparisons.
- ghost-sample: Compose demo app.

### 3. ghost-api — Annotations

Package: com.ghost.serialization.annotations. No external dependencies. Ten public annotations:

| Annotation                       | Target                       | Role                                       |
|----------------------------------|------------------------------|--------------------------------------------|
| <code>---</code>                 | <code>---</code>             | <code>---</code>                           |
| <code>@GhostSerialization</code> | <code>class</code>           | Marks model; discriminator, inferred, name |
| <code>@GhostName</code>          | <code>property</code>        | JSON alias                                 |
| <code>@GhostIgnore</code>        | <code>property</code>        | Omit from JSON                             |
| <code>@GhostEncoder</code>       | <code>property</code>        | Custom encode function                     |
| <code>@GhostDecoder</code>       | <code>property</code>        | Custom decode function                     |
| <code>@GhostFlatten</code>       | <code>property</code>        | Nested JSON → flat property                |
| <code>@GhostWrap</code>          | <code>property</code>        | Properties → JSON sub-object               |
| <code>@GhostResilient</code>     | <code>class/property</code>  | Tolerance for invalid types                |
| <code>@GhostFallback</code>      | <code>sealed subclass</code> | Default branch                             |
| <code>@GhostSignature</code>     | <code>sealed subclass</code> | Signature for inferred polymorphism        |

#### @GhostSerialization (required on models)

Without this annotation on the class, KSP ignores the type. It is the “contract” that you want codegen.

```
@GhostSerialization
data class Product(val id: Long, val title: String)
```

Optional parameters:

| Parameter            | Default           | Meaning                                                       |
|----------------------|-------------------|---------------------------------------------------------------|
| <code>---</code>     | <code>---</code>  | <code>---</code>                                              |
| <b>discriminator</b> | <b>"type"</b>     | JSON field name indicating subclass in sealed class           |
| <b>inferred</b>      | <b>false</b>      | If true, deduces subclass from present fields (no type field) |
| <b>name</b>          | <b>class name</b> | Registry alias if typeName collides                           |

Sealed with discriminator (most common case):

```
@GhostSerialization(discriminator = "kind")
sealed class Event {
    @GhostSerialization
    data class Click(val x: Int, val y: Int) : Event()
    @GhostSerialization
    data class Scroll(val delta: Int) : Event()
}
```

## Input JSON:

```
{"kind": "Click", "x": 10, "y": 20}
```

The generated serializer reads `kind`, does when, and calls the `Click` deserializer.

## Sealed inferred (APIs without type field):

```
@GhostSerialization(inferred = true)
sealed class SmartEvent { /* subclasses with unique fields */ }
```

If only `Click` has `x` and `y`, and `Scroll` only `delta`, generated code checks which fields appear in JSON and picks the subclass. More generated code, but tolerates legacy APIs.

## @GhostName — JSON name ≠ Kotlin name

```
@GhostSerialization
data class ApiUser(
    @GhostName("user_id") val id: Int,
    val name: String
)
```

- In Kotlin you use `id`.
- In JSON the field is called `user_id`.
- The generated serializer maps one to the other; you do not need Gson's `@SerializedName`.

## @GhostIgnore — local-only field, never in JSON

```
@GhostSerialization
data class Session(
    val token: String,
    @GhostIgnore val cachedAt: Long = System.currentTimeMillis()
)
```

On serialize, `cachedAt` does not appear. On deserialize, Kotlin's default is used. Useful for derived or UI fields.

## @GhostFlatten — nested JSON, flat property

External API sends:

```
{"settings": {"theme": "dark", "lang": "es"}}
```

Your model wants:

```
@GhostSerialization
data class Config(
    @GhostFlatten("settings.theme") val theme: String,
    @GhostFlatten("settings.lang") val lang: String
)
```

Ghost “flattens” the path when reading/writing. Several properties can point to different paths in the same subtree.

## @GhostWrap — several properties → one JSON sub-object

Inverse of flatten. If you have `val a: String`, `val b: Int` and want:

```
{"meta": {"a": "x", "b": 1}}
```

```
@GhostWrap("meta") val a: String // inside meta in JSON
```

## **@GhostEncoder / @GhostDecoder — types Ghost does not know**

For Instant, UUID, types from another library, you define static functions:

```
object DateCoder {
    fun decode(epoch: Long): Instant = Instant.ofEpochSecond(epoch)
    fun encode(instant: Instant): Long = instant.epochSecond
}
```

```
@GhostSerialization
```

```
data class Log(
    @GhostDecoder(provider = DateCoder::class, function = "decode")
    @GhostEncoder(provider = DateCoder::class, function = "encode")
    val created: Instant
)
```

In JSON created will be a number (epoch). The generated serializer calls `DateCoder.decode` instead of `nextString()`.

## **@GhostResilient — do not break on a rare enum**

If the backend sends `"role": "SUPERADMIN"` and your enum only has `USER`, `ADMIN`:

- Without resilient: exception.
- With `@GhostResilient` on the enum or property: null, default value, or skip depending on what KSP generates.

## **@GhostFallback — sealed with “catch-all”**

```
@GhostSerialization
```

```
sealed class Payment {
    @GhostSerialization @GhostFallback
    data class Unknown(override val rawType: String) : Payment()
}
```

If the discriminator does not match any known subclass, it deserializes to `Unknown` instead of failing. Ideal for APIs that add new types without notice.

## **@GhostSignature — inferred on sealed subclasses**

Marks which fields uniquely identify a subclass in inferred polymorphism. The compiler builds a decision tree; without this, inferred on complex sealed types may fail analysis.

## PART III — KSP compiler (internals)

### 4. KSP — Compiler entry point

KSP (Kotlin Symbol Processing) runs during compilation, in rounds. Ghost registers its processor via SPI:

META-INF/services/com.google.devtools.ksp.processing.SymbolProcessorProvider → class GhostSerializationProvider.

#### What happens when you run ./gradlew build

Step 1 — Discovery: KSP finds all classes annotated with @GhostSerialization in the module.

Step 2 — Analysis (GhostAnalyzer): for each class it validates types, names, flatten, sealed, defaults. If something is invalid, compile error with a message (no silent failure in 1.1.17).

Step 3 — Generation (GhostCodeGenerator + emitters): writes to disk, for example:

build/generated/ksp/<variant>/kotlin/<package>/UserSerializer.kt

Step 4 — Global registry: if there was at least one class, it also generates:

- GhostModuleRegistry\_my\_app.kt — clazz → serializer map
- META-INF/services/com.ghost.serialization.contract.GhostRegistry — for ServiceLoader on JVM
- META-INF/proguard/ghost-serialization.pro — R8 rules

Step 5 — Normal Kotlin compilation: the Kotlin compiler compiles your sources and generated ones together.

#### Consumer configuration

```
plugins {
    id("com.ghostserializers.ghost") version "1.2.2"
}

// Optional but recommended with several modules containing models:
ksp {
    arg("ghost.moduleName", "my_app") // → GhostModuleRegistry_my_app
}
```

If you omit moduleName, the registry is called GhostModuleRegistry\_Default. In tests (src/test) a \_Test suffix is added to avoid clashing with production.

#### Where to look at generated code

| Platform | Typical path under build/generated/ksp/  |
|----------|------------------------------------------|
| ---      | ---                                      |
| Android  | build/generated/ksp/androidDebug/kotlin/ |



|            |                                                        |
|------------|--------------------------------------------------------|
| <b>JVM</b> | <b>build/generated/ksp/main/kotlin/</b>                |
| <b>KMP</b> | <b>build/generated/ksp/metadata/commonMain/kotlin/</b> |

Open a `*Serializer.kt` and compare it with PART VIII of this manual (BenchUser walkthrough).

## 5. GhostAnalyzer and property models

`GhostAnalyzer.kt` is the compiler “notary”: it turns each `KSClassDeclaration` into a list of `GhostPropertyModel` that emitters consume.

### Validations it performs (and why they matter)

| Validation                                                                               | If it fails                                                     |
|------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| <code>!--</code>                                                                         | <code>!--</code>                                                |
| Only data class, sealed, enum, value class                                               | Does not generate code for bare interfaces                      |
| Supported types (primitives, String, List, Map nested <code>@GhostSerialization</code> ) | Error: cannot serialize <code>Flow&lt;T&gt;</code> or raw types |
| Coherent <code>@GhostFlatten</code> / <code>@GhostWrap</code> paths                      | Avoids JSON impossible to read/write                            |
| Unique JSON names per class                                                              | Two fields cannot map to the same key                           |
| Sealed: subclasses also annotated                                                        | Without annotation on child, not in registry                    |
| Custom coder: correct function signature                                                 | decode must be static and types compatible                      |

### What `GhostPropertyModel` stores (per-field metadata)

Each data class property is summarized in an internal object with:

- `kotlinName` / `jsonName` — may differ via `@GhostName`
- `nullable`, `hasDefault`, default value for multi-branch
- `bitIndex` — position in bit mask (field id → bit 0, name → bit 1, ...)
- flags: resilient, flatten path, wrap path, uses external encoder/decoder
- resolved type: enum, sealed, List of X, etc.

That metadata is what `DeserializeCodeEmitter` uses to write when (`selectNameAndConsume`) and constructor branches.

### Compile-time errors (1.1.17 behavior)

If the analyzer detects a problem, KSP reports an error and the build fails. It does not generate an empty serializer or “best effort”. This protects production: if it compiles, the model was coherent at compile-time.

## 6. Code generation — Emitters

| Class                         | Role                                        |
|-------------------------------|---------------------------------------------|
| <b>GhostCodeGenerator</b>     | <b>Orchestrates serialize + deserialize</b> |
| <b>SerializeCodeEmitter</b>   | <b>Serialize body (flat + streaming)</b>    |
| <b>DeserializeCodeEmitter</b> | <b>Deserialize body</b>                     |
| <b>StandardEmitter</b>        | <b>Normal models</b>                        |

**FragmentedEmitter****Huge models (>40 fields)****FragmentedSerializeEmitter****Avoids JVM method > 64KB**

Constants: `GhostEmitterConstants.kt` — %L, %T templates, registry names, chunk size 500 for registry when sharding.

**Generate serialize pattern (conceptual)**

// Generated — simplified idea

```
object UserSerializer : GhostSerializer<User> {
    override val typeName = "User"
    override fun serialize(writer: GhostJsonFlatWriter, value: User) {
        writer.beginObject()
        writer.writeNameHeader(HEADERS[0]) // "id":
        writer.writeInt(value.id)
        writer.writeNameHeader(HEADERS[1]) // "name":
        writer.writeString(value.name)
        writer.endObject()
    }
    override fun deserialize(reader: GhostJsonFlatReader): User {
        reader.beginObject()
        var mask = 0L
        var id = 0
        var name: String? = null
        loop@ while (reader.hasNext()) {
            when (reader.selectNameAndConsume(OPTIONS)) {
                0 -> { id = reader.nextInt(); mask = mask or 1L }
                1 -> { name = reader.nextString(); mask = mask or 2L }
                else -> reader.skipValue()
            }
        }
        reader.endObject()
        if ((mask and REQUIRED) != REQUIRED) throw GhostJsonException("Required field missing")
        return User(id, name!!)
    }
}
```

**selectNameAndConsume — O(1)**

`JsonReaderOptions` precomputes a trie of field names. At runtime there is no `== "fieldName"` string compare in the main loop hot path.

**Bitmask for required fields**

Imagine `User` with required fields `id`, `name`, `email`. The generator assigns:

- `id` → bit 1 (value 1)
- `name` → bit 2 (value 2)
- `email` → bit 4 (value 4)
- `REQUIRED_MASK` = 7 (1+2+4)

At the end of the loop, if `(mask and 7) != 7`, some field is missing and `GhostJsonException` is thrown with message "Required field 'email' missing". A single `Long` checks all required fields; cheaper than three if `(name == null)`.

## FragmentedEmitter (>40 fields)

JVM limits bytecode size per method (~64 KB). A `GodObject` with 80 fields would produce a huge deserialize. `FragmentedEmitter` splits the method into several `deserializePart1`, `deserializePart2`, etc., transparent to callers.

## 7. Registry and discovery

### Registry sharding (internal)

If a module has more than 500 models (`REGISTRY_CHUNK_SIZE`), the compiler generates:

- Several methods `getShardMap0()`, `getShardMap1()`, ...
- `getSerializer(clazz)` as a fragmented when to avoid exceeding 64 KB JVM bytecode per method
- lazy `allSerializers` for `prewarm()`

Generated file:  
`com.ghost.serialization.generated.GhostModuleRegistry_<moduleName>`  
 GhostRegistry interface:

```
interface GhostRegistry {
    fun <T : Any> getSerializer(clazz: KClass<T>): GhostSerializer<T>?
    fun getAllSerializers(): Map<KClass<*>, GhostSerializer<*>>
}
```

Generated: `GhostModuleRegistry_my_app` with `when (clazz)` possibly sharded if there are thousands of models (>500 per method).

JVM/Android discovery:

```
// Ghost.jvm.kt - idea
fun discoverRegistries(): Iterable<GhostRegistry> {
    // Class.forName fast-path + ServiceLoader
}
```

Generated file:  
`META-INF/services/com.ghost.serialization.contract.GhostRegistry`  
 iOS / Kotlin Native: `discoverRegistries()` returns an empty list (no `ServiceLoader`). You must register the module at startup:

```
// In the KMP framework init exported to iOS
fun initGhostRuntime() {
    Ghost.addRegistry(GhostModuleRegistry_my_app.INSTANCE)
    Ghost.prewarm()
}
```

```
}
```

Without this, `Ghost.deserialize` finds no serializer even though KSP generated the code.

Android/JVM: with `ServiceLoader` + generated `META-INF`, you usually do not need `addRegistry` except for tests or extra manual registries.

### addRegistry flow

When you call `addRegistry`, Ghost walks `registry.getAllSerializers()` and preloads `serializerCache` and `serializerByName`. Call it once at startup, not per request.

## 8. Ghost.kt — Public runtime API

Central file:  
ghost-serialization/src/commonMain/kotlin/com/ghost/serialization/Ghost.kt

### Full public API table

| Method                                                      | Description                                   |
|-------------------------------------------------------------|-----------------------------------------------|
| <code>---</code>                                            | <code>---</code>                              |
| <code>deserialize&lt;T&gt;(json: String)</code>             | Decode UTF-8 → flat reader → serializer       |
| <code>deserialize&lt;T&gt;(bytes: ByteArray)</code>         | No intermediate String copy                   |
| <code>deserialize&lt;T&gt;(source: BufferedSource)</code>   | Okio streaming                                |
| <code>deserialize(reader) / deserialize(flatReader)</code>  | Direct use in generated code                  |
| <code>encodeToString&lt;T&gt;(value)</code>                 | Flat writer → String                          |
| <code>encodeToBytes&lt;T&gt;(value)</code>                  | Flat writer → ByteArray                       |
| <code>encodeAndDiscard&lt;T&gt;(value)</code>               | Warm-up without result alloc                  |
| <code>serialize(sink, value)</code>                         | Bulk drain to Okio                            |
| <code>decodeFromBytes(bytes, KClass)</code>                 | Spring / Retrofit                             |
| <code>decodeFromSource(source, KClass)</code>               | Streaming + KClass                            |
| <code>encodeToSink(sink, value, KClass)</code>              | Encode with dynamic class                     |
| <code>getSerializer(KClass) / getSerializer(KType)</code>   | Resolution + cache                            |
| <code>addRegistry(registry)</code>                          | Required on iOS                               |
| <code>prewarm()</code>                                      | Loads registries + warmUp serializers         |
| <code>getSerializerByName / getSerializerNames</code>       | Compiler bridge                               |
| <code>throwError(msg)</code>                                | Uniform <code>IllegalArgumentException</code> |
| Constants <code>NOT_FOUND</code> , <code>MISSING_ANN</code> | Error messages                                |

Resolution order in `getSerializerFromRegistries`:

1. `mutableRegistries` (manual `addRegistry`)
2. `discoverRegistries()` — `ServiceLoader` JVM / empty on iOS
3. `registry.getSerializer(clazz)` — generated sharded when

### Resolving serializer

Order:

1. `serializerCache[KClass]` (primitives + prewarm)
2. `mutableRegistries` (manual `addRegistry`)
3. `discoverRegistries()` (`ServiceLoader` JVM)
4. `registry.getSerializer(clazz)` → generated when

## Deserialize — when to use each overload

```
// Most common: HTTP response already as String
val user = Ghost.deserialize<User>(responseBody)

// Better if you already have bytes (Retrofit/OkHttp without intermediate String)
val user = Ghost.deserialize<User>(bytes)

// Large streaming from Okio
val user = Ghost.deserialize<User>(bufferedSource)

// Advanced reader options (coercion, etc.) – requires internal OptIn
Ghost.deserialize<User>(json) { reader ->
    reader.coerceBooleans = true
}
```

Inside: `resolveSerializer<User>()` → `ghostInternalUseFlatReader` takes reader from pool → `UserSerializer.deserialize(flatReader)` → returns object.

## Encode — when to use each overload

```
val s = Ghost.encodeToString(order)    // UI, logs, SharedPreferences
val b = Ghost.encodeToBytes(order)     // send over network without String
Ghost.serialize(okioSink, order)       // write directly to socket/file
Ghost.encodeAndDiscard(order)          // only warm JIT, discard bytes
```

`serialize(value)` without sink is an alias of `encodeToString(value)` for compatibility.

## getSerializer — framework integration

Spring and Retrofit do not always have reified T. They use:

```
Ghost.getSerializer(Ordere::class)
Ghost.decodeFromBytes(bytes, Ordere::class)
```

For `List<Order>` in Retrofit, the factory resolves `ParameterizedType` and builds `ListSerializer(OrderSerializer)`.

## Prewarm — why and when

```
// Application.onCreate (Android) or @PostConstruct (Spring)
Ghost.addRegistry(GhostModuleRegistry_my_app.INSTANCE)    // only iOS/KMP without
ServiceLoader
Ghost.prewarm()
```

`prewarm()` does three things: discovers registries (JVM), fills `serializerCache` with all module serializers, and calls `warmUp()` on each

(dummy deserialize for JIT). The first real request pays less JIT compilation latency.

## 9. GhostSerializer — Dual contract

Every generated serializer implements `GhostSerializer<T>` in `contract/GhostSerializer.kt`.

### Why “dual” (two writers and two readers)

Ghost optimizes two distinct scenarios:

1. In-memory / HTTP body as bytes → `GhostJsonFlatWriter` + `GhostJsonFlatReader` on contiguous `ByteArray`. Maximum throughput.
2. Okio streaming (files, chunks) → `GhostJsonWriter` + `GhostJsonReader` on `BufferedSource`.

Generated code implements all four functions. Defaults on the interface may delegate flat ↔ streaming so you do not duplicate logic manually for simple types.

### Methods each FooSerializer implements

| Method                                         | Main use                                       |
|------------------------------------------------|------------------------------------------------|
| <code>----</code>                              | <code>----</code>                              |
| <code>serialize(GhostJsonFlatWriter, T)</code> | <code>encodeToString</code> , Retrofit request |
| <code>serialize(GhostJsonWriter, T)</code>     | <code>serialize(sink)</code>                   |
| <code>deserialize(GhostJsonFlatReader)</code>  | <code>Ghost.deserialize(bytes)</code>          |
| <code>deserialize(GhostJsonReader)</code>      | streaming + options                            |
| <code>warmUp()</code>                          | called from <code>prewarm()</code>             |
| <code>typeName</code>                          | readable key ("BenchUser")                     |
| <code>isResilient</code>                       | lists that skip bad elements                   |

### Mental example

```
Ghost.encodeToString(user)
    → BenchUserSerializer.serialize(flatWriter, user)
```

```
Retrofit response
    → BenchUserSerializer.deserialize(flatReader)
```

Always the same `BenchUserSerializer` object; different paths depending on buffer type.

## PART IV — Internal runtime

### End-to-end deserialize and encode pipeline

#### Deserialize (hot path)

JSON bytes

```

→ ghostInternalUseFlatReader (ThreadLocal pool)
→ GhostJsonFlatReader.reset(data)
→ FooSerializer.deserialize(reader)
    → beginObject()
    → loop: selectNameAndConsume(OPTIONS) // O(1) trie
    → nextInt / nextString / skipValue
    → bitmask REQUIRED_MASK
    → primary constructor (multi-branch if applicable)
→ T

```

## Encode (hot path)

```

value T
→ Ghost.resolveSerializer<T>()
→ ghostInternalEncodeWithWriter (WriterSinkPair pool)
→ FooSerializer.serialize(flatWriter, T)
    → pre-encoded writeNameHeader
    → writeInt / writeQuotedString
    → endObject()
→ FlatByteArrayWriter.toByteArray()
→ reset() — trims buffer if > maxWarmWriteBufferCapacity

```

## Why two writers and two readers

| Type                       | When                                            | Cost                             |
|----------------------------|-------------------------------------------------|----------------------------------|
| <b>GhostJsonFlatReader</b> | <b>Ghost.deserialize, Retrofit, Spring body</b> | <b>Minimum: contiguous array</b> |
| <b>GhostJsonReader</b>     | <b>Okio stream, coerceBooleans options</b>      | <b>More flexible</b>             |
| <b>GhostJsonFlatWriter</b> | <b>encodeToString/Bytes</b>                     | <b>Monomorphic hot path</b>      |
| <b>GhostJsonWriter</b>     | <b>serialize(sink)</b>                          | <b>Single Okio drain at end</b>  |

## 10. GhostJsonFlatReader — Fast parser

GhostJsonFlatReader is the parser used in almost the entire hot path of Ghost. It reads a ByteArray start to end with a position index, without allocating objects per token.

### Lifecycle in a typical deserialize

1. reset(bytes) — points at the response buffer (may be the same array from the pool).
2. beginObject() — expects {, prepares object context.
3. Loop hasNext() / selectNameAndConsume(OPTIONS) — identifies current field.
4. nextInt(), nextString(), etc. — consume value and advance position.
5. skipValue() — if JSON has fields your model does not (forward compatible).
6. endObject() — validates }.

7. Serializer checks mask and builds the data class.

### **selectNameAndConsume explained without jargon**

Instead of comparing strings each iteration (if (name == "email")), the compiler precomputes `JsonReaderOptions` with a mini-hash of field names. At runtime:

- Returns 0 if the field is "id"
- Returns 1 if it is "name"
- Returns -1 if the object ended
- Returns -2 if unknown field → typically `skipValue()`

That is amortized  $O(1)$  and JIT-friendly (fewer branches with strings).

### **Security limits**

| Parameter                | What it protects                  | Typical value              |
|--------------------------|-----------------------------------|----------------------------|
| ---                      | ---                               | ---                        |
| <b>maxDepth</b>          | <b>Infinite nested JSON [!!!]</b> | <b>255</b>                 |
| <b>maxCollectionSize</b> | <b>Huge lists/maps (DoS)</b>      | <b>50k Android, 1M JVM</b> |

If exceeded, clear exception. Does not limit total HTTP body size — configure that in `OkHttp/Spring`.

### **Typical generated code**

```
reader.beginObject()
while (true) {
    when (reader.selectNameAndConsume(OPTIONS)) {
        0 -> id = reader.nextInt()
        1 -> name = reader.nextString()
        -1 -> break
        -2 -> reader.skipValue()
    }
}
reader.endObject()
```

## **11. GhostJsonReader — Streaming parser**

parser/GhostJsonReader.kt — on Okio / streaming.

- String pool for repeated names
- `GhostDiscriminatorPeeker` for standard sealed
- Used when API requests `GhostJsonReader` in options lambda

Pools: `ThreadLocal` JVM/Android, `@ThreadLocal` iOS in `Ghost.*.kt`.

## **12. Writers — FlatByteArrayWriter**

writer/FlatByteArrayWriter.kt + `GhostJsonFlatWriter.kt`

- Growing flat buffer



- `reset()` after each encode — releases capacity above `maxWarmWriteBufferCapacity`
- JVM: 8 MB warm cap; Android/Native: 4 MB; (Wasm sources not published)

```
// After encode, reset keeps buffer up to cap
fun reset() {
    // needsComma=false, depth=0
    // if capacity > maxWarm → shrink to 8KB initial
}
```

Pre-encoded headers: bulk write of "field": without UTF-8 encode per field on hot path.

GhostJsonWriter — Okio variant for `serialize(sink)`.

### 13. GhostPools and scratch buffers

GhostPools.kt — `expect/actual`

- `acquireScratchBuffer(size)` / `releaseScratchBuffer`
  - Size tiers to avoid alloc in `number→string` conversion and ASCII escape
- Instance pools (in `Ghost.jvm.kt` / `.android.kt` / `.ios.kt`):
- `GhostJsonFlatReader` + `ByteArrayGhostSource`
  - `GhostJsonReader`
  - `WriterSinkPair` (flat writer + sink)
- `runSynchronized` on registry caches.

### 14. GhostHeuristics — Platform limits

GhostHeuristics (`expect/actual` per target) defines compiled limits per platform. They are not flags the user changes in `Ghost.deserialize()` today.

#### Row-by-row table

| Property                 | Android       | iOS/Native     | JVM server       | Practical meaning                                                      |
|--------------------------|---------------|----------------|------------------|------------------------------------------------------------------------|
| ---                      | ---           | ---            | ---              | ---                                                                    |
| <b>maxCollectionSize</b> | <b>50,000</b> | <b>500,000</b> | <b>1,000,000</b> | <b>Max elements when parsing List or Map</b>                           |
| <b>maxWarmWriteBu</b>    | <b>4 MB</b>   | <b>4 MB</b>    | <b>8 MB</b>      | <b>After encode, if internal buffer exceeds this, shrinks to ~8 KB</b> |
| <b>maxDepth</b>          | <b>255</b>    | <b>255</b>     | <b>255</b>       | <b>Max nesting of { [</b>                                              |

`maxCollectionSize`: protects against `{"items":[...millions...]}`. On JVM server the limit is higher because legitimate batch APIs can be large; on

mobile it is stricter for RAM.

`maxWarmWriteBufferCapacity`: if you serialize a 6 MB JSON once, the buffer grows; after `reset()`, Ghost does not keep 6 MB forever on mobile (4 MB cap) but on JVM may keep up to 8 MB to avoid realloc on the next similar response. This improved server write benchmarks without inflating Android RAM.

HTTP body: a 50 MB POST can enter memory whole if OkHttp/Spring allow it; Ghost will parse until `maxCollectionSize` or system memory says stop. Configure HTTP limits outside Ghost.

## Collections supported at runtime (verified in code)

| Type                              | Support                                                                                                                                                                                                 |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>---</code>                  | <code>---</code>                                                                                                                                                                                        |
| <code>List&lt;T&gt;</code>        | Yes — <code>ListSerializer</code> via <code>getSerializer(KType)</code>                                                                                                                                 |
| <code>Map&lt;String, V&gt;</code> | Yes — <code>MapSerializer</code>                                                                                                                                                                        |
| <code>Set&lt;T&gt;</code>         | No <code>SetSerializer</code> in ghost-serialization; the README mentions it but the current runtime only resolves <code>List/Map</code> besides primitives and <code>@GhostSerialization</code> models |

## 15. Primitive and collection serializers

`serializers/PrimitiveSerializers.kt`, `ListSerializer`, `MapSerializer`, etc.

Ghost resolves `Int`, `String`, `List<T>`, `Map<String,V>` with built-in serializers before the registry.

For generics: `getSerializer(KType)` builds or caches `ListSerializer(element)`.

## 16. Sealed, inferred, multi-branch

### Sealed with discriminator (flow to visualize)

JSON arrives → sealed serializer reads type field (or what you define) → when picks sub-serializer → `ClickSerializer.deserialize` or `ScrollSerializer.deserialize`.

Advantage: explicit, easy to debug. Disadvantage: backend must always send the discriminator.

### Inferred (no type field)

The compiler analyzes which field combination is unique per subclass. Generates a tree of if/when like: “if x and y present → Click; if only delta → Scroll”. Single JSON pass.

Useful when migrating legacy APIs. More generated bytecode and more branches in the serializer.

## Multi-branch constructors (Kotlin defaults)

BenchUser has `isActive = true`, `role = VIEWER`, `bio = null`. They may be missing in JSON. Instead of:

```
// What Ghost does NOT want on hot path
```

```
BenchUser(id, name, email, score).copy(isActive = true, role = VIEWER)
```

The compiler generates up to  $2^N$  branches if `((mask and X) == X)` return `BenchUser(...)` (one constructor call per branch) instead of chaining `.copy()`. Fewer temporary objects, less GC.

## PART V — HTTP adapters

### 17. ghost-gradle-plugin

GhostPlugin.kt when applied:

1. If KSP → adds ghost-compiler to ksp / kspCommonMainMetadata
2. If KMP → ghost-serialization + ghost-api on commonMain
3. If Android/JVM → runtime implementation + api
4. afterEvaluate: if Retrofit/Ktor on classpath → ghost-retrofit / ghost-ktor

```
ghost {
    version.set("1.2.2")
    autoInjectKtor.set(true)
    autoInjectRetrofit.set(true)
}
```

Plugin id: `com.ghostserializer.ghost`. `DEFAULT_VERSION` in `plugin = 1.2.2`.

### 18. ghost-retrofit

#### Full configuration

```
dependencies {
    implementation("com.ghostserializer:ghost-retrofit:1.2.2")
}

interface ApiService {
    @GET("user/{id}")
    suspend fun getUser(@Path("id") id: Int): User // User must have @GhostSerialization

    @POST("users")
    suspend fun createUser(@Body user: User): User
}

val retrofit = Retrofit.Builder()
    .baseUrl("https://api.example.com/")
```

```
.addConverterFactory(GhostConverterFactory.create()) // before Gson if both exist
.client(okHttpClient)
.build()
```

## What the factory does internally

Response: reads `ResponseBody` into a reusable buffer (scratch), then `ghostInternalUseFlatReader` → `serializer.deserialize(flatReader)`. Does not go through intermediate UTF-16 String if the body is already bytes.

Request: `ghostInternalEncodeWithWriter { w -> serializer.serialize(w, value) }` → `RequestBody` with `application/json`.

Cache: `ConcurrentHashMap<Type, GhostSerializer>` — first call per type resolves; later calls are  $O(1)$ .

null body: if the Retrofit method allows nullable body and null arrives, converter returns null without serializing.

## Coexisting with Gson

If you add `GhostConverterFactory` before `GsonConverterFactory`, Retrofit tries Ghost first. Only types with a Ghost serializer use Ghost; the rest fall through to Gson.

## 19. ghost-ktor

`GhostContentConverter.kt` + `GhostKtorExtensions.kt`

```
val client = HttpClient {
    install(ContentNegotiation) {
        ghost() // Ktor 2.3.x
    }
}
```

Same pool + flat reader/writer pattern. Ktor 3 in consumer apps may need a custom converter (see android test app `GhostKtor3Converter`).

## 20. ghost-spring-boot-starter

### Dependencies

```
plugins {
    id("com.ghostserializer.ghost") version "1.2.2"
}
dependencies {
    implementation("com.ghostserializer:ghost-spring-boot-starter:1.2.2")
}
```

The ghost plugin in the same module as your `@RestController` and DTOs with `@GhostSerialization`.

### Auto-configuration (Boot 3.4+)

|       |      |         |          |     |
|-------|------|---------|----------|-----|
| Three | open | classes | imported | via |
|-------|------|---------|----------|-----|

META-INF/spring/org.springframework.boot.autoconfigure.AutoConfiguration.imports:

| Class                                | Effect                                                      |
|--------------------------------------|-------------------------------------------------------------|
| ---                                  | ---                                                         |
| <b>GhostAutoConfiguration</b>        | <b>Marker / imports</b>                                     |
| <b>GhostWebMvcAutoConfiguration</b>  | <b>Registers GhostHttpMessageConverter at start of list</b> |
| <b>GhostWebFluxAutoConfiguration</b> | <b>Reactive encoder/decoder codecs for WebFlux</b>          |

No ghost.\* in application.properties. Everything is automatic if the DTO has a serializer.

## How it interacts with Jackson

Spring Boot brings Jackson by default. Ghost inserts its converter at index 0:

- If `Ghost.getSerializer(Order::class) != null` → Ghost serializes/deserializes.
- If no Ghost serializer → Jackson continues with the DTO as before.

You can have in the same project: JPA entities with Jackson and API DTOs with Ghost.

## Typical controller (no special changes)

```
@RestController
@RequestMapping("/api")
class OrderController(private val service: OrderService) {

    @GetMapping("/orders/{id}")
    fun get(@PathVariable id: Long): OrderDto = service.find(id)

    @PostMapping("/orders")
    fun create(@RequestBody dto: OrderDto): OrderDto = service.create(dto)
}
```

```
@GhostSerialization
data class OrderDto(val id: Long, val items: List<LineItem>)
```

## Body size limit (important)

Ghost does not cut 100 MB bodies. Configure in Spring:

```
spring:
  codec:
    max-in-memory-size: 10MB
```

And/or limits in nginx/API gateway. See PART VII security model.  
Spring Boot 3.4.5 verified in starter tests.

## 21. ProGuard / R8

Generated: META-INF/proguard/ghost-serialization.pro

Consumer rules in ghost-api. Serializers and generated package must be kept if you use minify — the plugin emits rules.

In Android test app: keep @GhostSerialization and generated.\*\*.

## 22. Maven Central publishing

Publishable (publish.gradle.kts): ghost-\* except sample, benchmark, integration-test.

From Mac for full KMP (iosArm64 + iosSimulatorArm64 + JVM + Android AAR).

```
./gradlew publishToSonatype closeAndReleaseSonatypeStagingRepository
```

Coordinates: com.ghostserializer:\*:1.2.2

From Linux: iOS variants may be missing on Central.

# PART VI — CI, tests, and commands

## 23. CI and tests

Single workflow: .github/workflows/ci.yml

| Job           | Runner        | Command                                           | What it validates                                                |
|---------------|---------------|---------------------------------------------------|------------------------------------------------------------------|
| ---           | ---           | ---                                               | ---                                                              |
| test-jvm      | ubuntu-latest | ./gradlew ciTestJvm                               | Compiler,<br>integration-test, retrofit,<br>ktor, spring, plugin |
| test-android  | ubuntu-latest | :ghost-serialization:te                           | Android runtime                                                  |
| test-ios      | macos-14      | kspCommonMainKotlin<br>+<br>iosSimulatorArm64Test | K/N iOS; fails if SKIPPED                                        |
| publish-check | ubuntu-latest | publishToMavenLocal<br>-PskipTests                | Maven packaging                                                  |

iOS CI: Kotlin/Native does not print "N tests completed". The workflow trusts Gradle exit code and detects SKIPPED/FAILED in the log.

ciTestJvm in root build.gradle.kts:

- ghost-serialization jvmTest
- ghost-compiler test
- ghost-integration-test
- ghost-retrofit, ghost-ktor jvmTest
- ghost-spring-boot-starter test
- ghost-gradle-plugin test

GitHub jobs: JVM, Android testDebugUnitTest, iOS macos-14.

Verified on this repo (./gradlew ciTestJvm, May 2026): 416 JVM-module tests. Full ./gradlew ciTest on Linux: 642 (416 ciTestJvm + 226 testDebugUnitTest Android). On macOS with Xcode: ~874 (642 + ~232 iosSimulatorArm64Test per README).

ghost-benchmark:run depends on ciTest (except -PskipTests).

## Reference Gradle commands

| Goal                    | Command                                                                         |
|-------------------------|---------------------------------------------------------------------------------|
| ---                     | ---                                                                             |
| Local CI (JVM)          | ./gradlew ciTestJvm                                                             |
| Full CI (+ Android)     | ./gradlew ciTest                                                                |
| iOS (Mac only)          | ./gradlew :ghost-serialization:iosSimulatorArm64Test                            |
| View generated KSP code | ./gradlew :ghost-integration-test:compileKotlin                                 |
| Benchmark               | ./gradlew :ghost-benchmark:run --args="--runs 10000 --warmup 20000"             |
| Benchmark without tests | ./gradlew :ghost-benchmark:run -PskipTests --args="--runs 10000 --warmup 20000" |
| Sample Android          | ./gradlew :ghost-sample:assembleDebug                                           |
| Publish local           | ./gradlew publishToMavenLocal -PskipTests                                       |
| Maven Central           | ./gradlew publishToSonatype closeAndReleaseSonatypeStagingRepository            |
| Regenerate manual PDF   | .venv-pdf/bin/python scripts/build_ghost_manual_pdf.py                          |

Toolchain: JDK 17, Kotlin/KSP per gradle/libs.versions.toml.

## 24. Test apps (your 3 repos)

| App                        | What it validates                         |
|----------------------------|-------------------------------------------|
| ghost-android-test-app     | Gson/Moshi/KSer vs Ghost, Retrofit, Ktor3 |
| ghost-spring-boot-test-app | Jackson vs Ghost WebFlux, benchmark.py    |
| ghost-ios-test-app         | XCFramework + GhostBridge + Codable       |

All use 1.2.2 Maven Central (no mavenLocal in final config).

## 25. End-to-end flow on Android

### Step by step (from scratch)

1. settings.gradle.kts — pluginManagement { gradlePluginPortal() }
2. app/build.gradle.kts — id("com.ghostserializer.ghost") version "1.2.2"
3. Create data class with @GhostSerialization in the network package
4. Build → Make Project — verify UserSerializer.kt exists in app/build/generated/ksp/
5. Application.onCreate: Ghost.prewarm() (optional but recommended for

high-traffic apps)

6. Retrofit: `.addConverterFactory(GhostConverterFactory.create())`

7. Release: confirm ProGuard keep includes  
`com.ghost.serialization.generated.**`

## What happens on the first Retrofit request

OkHttp receives bytes

- `GhostConverterFactory`
- `ghostInternalUseFlatReader`
- `UserSerializer.deserialize`
- User object in memory

No reflection on that path.

## 26. End-to-end flow on iOS

1. KMP shared module with ghost plugin + export serialization/api
2. assembleXCFramework on Mac
3. Swift imports framework
4. `Ghost.addRegistry(INSTANCE)` in Kotlin bridge
5. deserialize via bridge — prefer UTF-8 String vs raw bytes

## 27. End-to-end Spring flow

1. implementation ghost-spring-boot-starter
2. ghost plugin + ksp in same module as controllers
3. Annotated DTOs
4. Normal controllers `@RequestBody` / `@ResponseBody`
5. Ghost converter only for types with serializer

# PART VII — Publishing, security, errors

## Security model and limits

Ghost separates three policy layers:

1. Parser (Ghost): `maxCollectionSize`, `maxDepth` — anti-DoS on lists/maps and nesting.
2. Encode memory: `maxWarmWriteBufferCapacity` — does not retain huge buffers between requests.
3. HTTP (your app): body size, timeouts, reverse proxy.

A 100 MB HTTP body is not rejected automatically by Ghost; configure `OkHttp`, `spring.codec.max-in-memory-size`, `nginx`.



GhostJsonException includes approximate position in the buffer to debug malformed JSON or missing required fields.

## 28. Common errors

### NOT\_FOUND or “No serializer found”

Typical message: includes NOT\_FOUND and the type’s KClass.

Causes and fix:

1. Class lacks @GhostSerialization → add annotation and recompile.
2. KSP did not run (broken clean, module without ghost plugin) → ./gradlew :app:kspDebugKotlin or rebuild.
3. Model in another module without KSP in that module → each module with models needs plugin + ksp.
4. 

iOS/KMP:

forgot

  
Ghost.addRegistry(GhostModuleRegistry\_xxx.INSTANCE) at framework startup.
5. R8 removed serializers → add generated.\*\* and @GhostSerialization rules.

### GhostJsonException — missing required field

Message: Required field 'email' missing in JSON

Means: JSON did not include a field that has no Kotlin default. Not a Ghost bug: API contract mismatch.

What to do: fix backend, make field nullable, or give Kotlin default.

### GhostJsonException — malformed JSON

Position in the message points to approximate byte of error (missing comma, unclosed string). Useful to log raw request body.

### maxCollectionSize exceeded

List or map in JSON exceeded platform limit (50k on Android). May be legitimate large JSON or DoS attack.

What to do: paginate API, or if you control the reader in custom decoder, adjust maxCollectionSize on that path (not on global pooled Ghost.deserialize).

### Spring still uses Jackson for my DTO

GhostHttpMessageConverter only handles classes where Ghost.getSerializer(clazz) != null. If your @RequestBody Data lacks @GhostSerialization in the same module compiled with KSP, Jackson wins.

Fix: annotate DTO, apply ghost plugin to controller module, verify Ghost

converter is registered (starter does index 0).

### Plugin `com.ghostserializer.ghost` not found

Gradle does not resolve the plugin. Check `pluginManagement` in `settings.gradle.kts` with `gradlePluginPortal()`, version 1.2.2 on Maven Central, and sync again.

### iOS: works in debug, fails in release

Almost always R8/ProGuard or missing `addRegistry`. Check generated rules in `META-INF/proguard/` of the module with KSP.

## 29. Quick API — cheat sheet

```
Ghost.deserialize<T>(json: String)
Ghost.deserialize<T>(bytes: ByteArray)
Ghost.encodeToString(value)
Ghost.encodeToBytes(value)
Ghost.serialize(sink, value) // Okio bulk write
Ghost.addRegistry(registry) // required on iOS
Ghost.prewarm()
Ghost.getSerializer(MyClass::class)
```

## 30. Key files to read in the repo

| Topic              | Path                                                                           |
|--------------------|--------------------------------------------------------------------------------|
| ----               | ----                                                                           |
| <b>API</b>         | <code>ghost-api/src/commonMain/kotlin/com/ghost/serialization</code>           |
| <b>Processor</b>   | <code>ghost-compiler/src/main/kotlin/com/ghost/serialization</code>            |
| <b>Generator</b>   | <code>ghost-compiler/src/main/kotlin/com/ghost/serialization</code>            |
| <b>Runtime</b>     | <code>ghost-serialization/src/commonMain/kotlin/com/ghost/serialization</code> |
| <b>Flat reader</b> | <code>ghost-serialization/src/commonMain/kotlin/com/ghost/serialization</code> |
| <b>Flat writer</b> | <code>ghost-serialization/src/commonMain/kotlin/com/ghost/serialization</code> |
| <b>JVM pools</b>   | <code>ghost-serialization/src/jvmMain/kotlin/com/ghost/serialization</code>    |
| <b>iOS</b>         | <code>ghost-serialization/src/iosMain/kotlin/com/ghost/serialization</code>    |
| <b>Retrofit</b>    | <code>ghost-retrofit/src/main/kotlin/com/ghost/serialization</code>            |
| <b>Ktor</b>        | <code>ghost-ktor/src/commonMain/kotlin/com/ghost/serialization</code>          |
| <b>Spring</b>      | <code>ghost-spring-boot-starter/src/main/kotlin/com/ghost/serialization</code> |
| <b>Plugin</b>      | <code>ghost-gradle-plugin/src/main/kotlin/com/ghost/gradle</code>              |

## 31. Version 1.2.2 — relevant changes

- KSP2 AST Scan & Compiler Optimizations: Integrated full KSP2 compatibility (aligned with Kotlin 2.1.10-1.0.31) for faster incremental builds and improved compile-time AST inspection.
- Perfect Hashing & O(1) Field Lookup: Optimized key dispatch tables utilizing precomputations of 4-byte hash slots inside `JsonReaderOptions`

to minimize collisions during field lookup.

- Streaming Key-Match Optimization: Specialized `verifyKeyMatch` loop inside `GhostJsonReader` to branch on `isStreaming` once at loop start, enabling direct `ByteArray` or `BufferedSource` reads, yielding a 2x throughput boost on binary payloads.
- Zero-Allocation Coercion: Handled string-to-boolean/number coercions natively at byte level, bypassing intermediary string allocations.
- Wiki & Badges Revamp: Updated documentation badges and flowcharts to rasterized `.png` format for clean out-of-the-box rendering inside Android Studio's Markdown preview.

---

### 31.5. Version 1.1.17 — legacy changes

- No `maxPayloadBytes` in core (HTTP limits in `OkHttp/Spring/nginx`)
  - JVM warm buffer 8 MB (fix inflated `WRITE` benchmark)
  - Spring split open `AutoConfiguration` (Boot 3.4)
  - Centralized `ciTestJvm`
  - `SpringBootTest` in starter
  - CI: 416 (`ciTestJvm`) / 642 (`ciTest` on Linux) — verified by Gradle test XML
- 

## 32. Glossary

- KSP: Kotlin Symbol Processing — compile-time codegen
  - Flat path: reader/writer on contiguous `ByteArray`
  - Registry: `clazz` → generated serializer map
  - Shard: split registry when for JVM limit
  - Trie options:  $O(1)$  structure for JSON field names
  - `ServiceLoader`: JVM discovery of `GhostRegistry`
- 

## 33. GhostSerializationProcessor — real code

File:

`ghost-compiler/src/main/kotlin/com/ghost/serialization/compiler/GhostSerializationProcessor.kt`

The processor is the heart of compile-time. On each KSP round:

```
override fun process(resolver: Resolver): List<KSAnnotated> {
    val symbols = resolver.getSymbolsWithAnnotation(
        C.STR_ANNOTATION_SERIALIZATION
    )
    val validClasses = symbols.filterIsInstance<KSClassDeclaration>().toList()
    validClasses.forEach { processClass(it) }
```

```

    if (validClasses.isNotEmpty()) {
        generateModuleRegistry()
        generateProGuardRules()
        generateServiceFile()
    }
    return unableToProcess.toList()
}

```

Registry      name:      KSP      option      ghost.moduleName      →  
 GhostModuleRegistry\_<name>. If code lives in src/test, \_Test suffix is  
 added to avoid colliding with production registry.

Per annotated class outputs:

- com.your.package.UserSerializer (object implementing GhostSerializer<User>)
- Entry in internal registry map

Global outputs (once per module):

- GhostModuleRegistry\_my\_app.kt with INSTANCE
- META-INF/services/com.ghost.serialization.contract.GhostRegistry
- META-INF/proguard/ghost-serialization.pro

## 34. JsonReaderOptions — O(1) trie in detail

parser/JsonReaderOptions.kt precomputes a 1024-slot dispatch table.

```

class JsonReaderOptions{
    internal val rawBytes: Array<ByteArray>,
    internal val shift: Int,
    internal val multiplier: Int
} {
    internal val dispatch = IntArray(1024) { -1 }
    // init: hash of up to 4 bytes of field name → index
}

// In generated serializer:
private val OPTIONS = JsonReaderOptions.of(
    0, 31, "id", "name", "email"
)

```

In the deserialize loop:

```

when (reader.selectNameAndConsume(OPTIONS)) {
    0 -> id = reader.nextInt()
    1 -> name = reader.nextString()
    -1 -> break      // end of object
    -2 -> reader.skipValue() // unknown field
}

```

Why it matters: comparing strings on every field of a large JSON destroys

performance. Ghost compares buffer bytes with names precomputed in `rawBytes`.

## 35. Ghost.jvm.kt — pools and discovery

ThreadLocal pools (same idea on Android):

```
private val flatReaderPool = ThreadLocal<GhostJsonFlatReader>()
private val writerPool = ThreadLocal<WriterSinkPair>()

actual fun ghostInternalEncodeWithWriter(
    block: (GhostJsonFlatWriter) -> Unit
): ByteArray {
    val pair = acquireFlatWriterPair()
    block(pair.writer)
    val result = pair.byteWriter.toByteArray()
    pair.byteWriter.reset()
    return result
}
```

discoverRegistries() on JVM:

1.

`Class.forName("com.ghost.serialization.generated.GhostModuleRegistry_Default").getField("INSTANCE")` (JVM fast path)

2. `ServiceLoader.load(GhostRegistry::class.java)` reads META-INF/services  
That is why on Android/JVM you do not need `addRegistry` if KSP generated the service file and R8 did not remove it.

## 36. Ghost.ios.kt — critical differences

```
actual fun discoverRegistries(): Iterable<GhostRegistry> = emptyList()
```

iOS/Native does not use `ServiceLoader`. You must register manually at KMP framework startup:

```
// shared/src/iosMain or bridge
fun initGhost() {
    Ghost.addRegistry(GhostModuleRegistry_my_app.INSTANCE)
    Ghost.prewarm()
}
```

Pools with `@ThreadLocal` instead of `java.lang.ThreadLocal`.

Synchronization with `objc_sync_enter/exit` for shared caches.

Framework export: the module with models + KSP must export `ghost-serialization` and `ghost-api` in `XCFramework export(...)`.

## 37. GhostConverterFactory — Retrofit flow

```

class GhostConverterFactory private constructor() : Converter.Factory() {

    override fun responseBodyConverter(type, annotations, retrofit):
Converter<ResponseBody, *>? {
        val serializer = getSerializerWithCache(type) ?: return null
        return Converter { body ->
            body.use {
                // Read stream → scratch buffer → flat reader
                ghostInternalUseFlatReader(scratch, offset) { reader ->
                    serializer.deserialize(reader)
                }
            }
        }
    }

    override fun requestBodyConverter(type, annotations, retrofit): Converter<*,
RequestBody>? {
        val serializer = getSerializerWithCache(type) ?: return null
        return Converter { value ->
            ghostInternalEncodeWithWriter { w ->
                serializer.serialize(w, value)
            }.toRequestBody(MEDIA_TYPE)
        }
    }
}

```

- `getSerializerWithCache` supports `List<T>` and `Map<String,V>` via `ParameterizedType`
- If no serializer → null → Retrofit tries another converter
- Media type: `application/json`

## 38. Ktor integration — `ghost()`

`GhostContentConverter.kt` implements Ktor's `ContentConverter`.

```

fun ContentNegotiation.Config.ghost() {
    register(ContentType.Application.Json, GhostContentConverter())
}

```

Same flat reader/writer path as Retrofit. The Gradle plugin can inject `ghost-ktor` automatically if it detects Ktor on classpath (`autoInjectKtor`).

Ktor 3: some apps use a different API; the android test app includes `GhostKtor3Converter` as reference if `ghost()` does not compile.

## 39. Spring — three auto-configurations

| Class                         | Activates               |
|-------------------------------|-------------------------|
| <b>GhostAutoConfiguration</b> | <b>Marker / imports</b> |

**GhostWebMvcAutoConfiguration****GhostHttpMessageConverter in MVC****GhostWebFluxAutoConfiguration****Reactive codecs****GhostHttpMessageConverter:**

```

override fun supports(clazz: Class<*>) =
    Ghost.getSerializer(clazz.kotlin) != null

override fun readInternal(clazz: Class<*>, inputMessage: HttpInputMessage) =
    Ghost.decodeFromBytes(inputMessage.body.readBytes(), clazz.kotlin)

override fun writeInternal(t: Any, outputMessage: HttpOutputMessage) {
    val ser = Ghost.getSerializer(t::class as KClass<Any>)!!
    val bytes = ghostInternalEncodeWithWriter { w ->
        ser.serialize(w, t)
    }
    outputMessage.body.write(bytes)
}

```

The converter is inserted at index 0 to take priority over Jackson on Ghost types.

## 40. integration-test sample models

Package:

ghost-integration-test/src/main/kotlin/com/ghost/serialization/integration/model/

Types to study in the repo (sources, not generated):

- BenchUser — typical user benchmark
- ComplexObject — deep nesting
- SmartEvent — sealed inferred
- DeviceCommand — sealed with discriminator
- HugeModel — >40 fields → FragmentedEmitter
- DeepNestedModel — JSON depth

After ./gradlew :ghost-integration-test:compileKotlin, open:

ghost-integration-test/build/generated/ksp/main/kotlin/

There you will see real \*Serializer.kt and GhostModuleRegistry\_Default.kt.

## 41. FragmentedEmitter — huge models

When a data class exceeds ~40 properties, the compiler splits serialize/deserialize into fragments to avoid the 64 KB JVM bytecode limit per method.

Symptom if it did not exist: compile error on generated serializer for

massive models (legacy APIs, ERP DTOs).

## 42. Multi-branch constructors ( $\leq 4$ properties with default (MAX\_DEFAULT\_BRANCH\_COUNT = 4))

If a constructor has up to 3 parameters with default, KSP can generate when (mask) branches that call the primary constructor once with the correct argument combination, avoiding:

```
// Anti-pattern Ghost avoids on hot path
User(id).copy(name = n)
```

This reduces allocations on deserialize of objects with many optional fields.

## 43. Sealed inferred vs discriminator

With discriminator (discriminator = "type"):

```
{"type": "click", "x": 10, "y": 20}
```

Inferred (inferred = true): compiler generates decision tree from which fields appear:

```
{"x": 10, "y": 20}
```

No type field. Useful for inconsistent APIs; more generated code and more serializer branches.

## 44. Resilient and Fallback at runtime

@GhostResilient on enum: unknown JSON value → null or first value per generated policy.

@GhostFallback on sealed: default subclass if discriminator does not match.

GhostSerializer.isResilient: lists that tolerate malformed elements (skip instead of abort).

## 45. ghostInternalUseFlatReader — decode pipeline

```
inline fun <T> ghostInternalUseFlatReader(
    bytes: ByteArray,
    limit: Int = bytes.size,
    block: (GhostJsonFlatReader) -> T
): T {
    // Acquire reader from pool, reset(bytes), run block, return
}
```

Ghost.deserialize<T>(json) converts String → UTF-8 bytes → flat reader → serializer.deserialize(reader).



`Ghost.deserialize<T>(bytes)` skips String conversion if you already have bytes (Retrofit, OkHttp).

## 46. encodeToString vs serialize(sink)

| API                          | Output            | Use                                                     |
|------------------------------|-------------------|---------------------------------------------------------|
| <code>encodeToString</code>  | UTF-8 String      | UI, logs, tests                                         |
| <code>encodeToBytes</code>   | ByteArray         | network, binary cache                                   |
| <code>serialize(sink)</code> | Okio BufferedSink | streaming to socket/file without full intermediate copy |

All use `GhostJsonFlatWriter` + `FlatByteArrayWriter` internally on hot path.

## 47. Gradle — dependencies the plugin injects

KMP commonMain:

- `com.ghostserializer:ghost-serialization`
- `com.ghostserializer:ghost-api (api)`

KSP:

- `com.ghostserializer:ghost-compiler` on `ksp` / `kspCommonMainMetadata` / `targets`

Conditional afterEvaluate:

- Retrofit on classpath → `ghost-retrofit`
- Ktor on classpath → `ghost-ktor`

```
ghost {
    version.set("1.2.2") // or omit if plugin brings DEFAULT_VERSION
}
```

## 48. R8 / ProGuard — generated rules

The processor writes rules that keep:

- Generated `*Serializer` classes
- `GhostModuleRegistry_*` and its `INSTANCE` field
- `GhostRegistry` interfaces for `ServiceLoader`

On Android consumer:

```
-keep @com.ghost.serialization.annotations.GhostSerialization class * { *; }
-keep class com.ghost.serialization.generated.** { *; }
```

Without this: `NOT_FOUND` in release even if debug works.

## 49. Benchmark and CI — useful commands

```
# Monorepo JVM tests (like CI)
./gradlew ciTestJvm
```

```
# Benchmark (requires prior tests except -PskipTests)
./gradlew :ghost-benchmark:run

# Regenerate serializers locally
./gradlew :ghost-integration-test:compileKotlin

# PDF of this manual
.venv-pdf/bin/python scripts/build_ghost_manual_pdf.py
```

PDF output: docs/Ghost-Serialization-Manual-1.2.2.pdf (A5 format).

## 50. Study checklist (40 min on phone)

- 1. Read ch. 1-4 (philosophy + annotations)
- 2. Read ch. 6-7 (generated serializer + registry)
- 3. Read ch. 9-12 (Ghost API + flat reader/writer)
- 4. Read ch. 35-36 (JVM vs iOS — critical)
- 5. Adapter for your stack: 37 Retrofit, 38 Ktor, 39 Spring
- 6. On PC open a generated \*Serializer.kt and compare with ch. 6
- 7. Review ch. 28 common errors before production integration

# PART VIII — integration-test catalog and generated code

Module :ghost-integration-test is the compiler and runtime regression lab. After ./gradlew :ghost-integration-test:compileKotlin, inspect: ghost-integration-test/build/generated/ksp/main/kotlin/

## VIII.1 Full model catalog

96 types annotated with @GhostSerialization in src/main + src/test (includes sealed subclasses). 26 test classes in src/test/kotlin/com/ghost/serialization/integration/.

### A. Benchmark and real API shape

| Model            | File                | What it validates                      |
|------------------|---------------------|----------------------------------------|
| ----             | ----                | ----                                   |
| BenchUser        | BenchUser.kt        | Multi-branch (3 defaults), nested enum |
| BenchResult      | BenchResult.kt      | Benchmark response                     |
| BenchmarkMetrics | BenchmarkMetrics.kt | Aggregated metrics                     |
| ComplexResponse  | ComplexResponse.kt  | Composite API payload                  |
| ApiUserEvent     | ApiUserEvent.kt     | User event                             |

**ApiProductConfig**

## **ApiProductConfig.kt**

**Product config**

|                        |                           |                                      |
|------------------------|---------------------------|--------------------------------------|
| <b>ExtremeMetadata</b> | <b>ExtremeMetadata.kt</b> | <b>Bulky metadata</b>                |
| <b>Address</b>         | <b>Address.kt</b>         | <b>Simple nesting</b>                |
| <b>Category</b>        | <b>Category.kt</b>        | <b>Catalog</b>                       |
| <b>Tag</b>             | <b>Tag.kt</b>             | <b>Tags</b>                          |
| <b>Priority</b>        | <b>Priority.kt</b>        | <b>Priority enum</b>                 |
| <b>UserRole</b>        | <b>UserRole.kt</b>        | <b>Role enum (used by BenchUser)</b> |

Test: GhostStressTest, GhostLibraryMethodTest

## B. Nesting, graphs, and generics

| Model                     | What it validates               |
|---------------------------|---------------------------------|
| <b>----</b>               | <b>----</b>                     |
| <b>ComplexObject</b>      | <b>Deep multi-level object</b>  |
| <b>NestedContainer</b>    | <b>Nested container</b>         |
| <b>RecursiveNode</b>      | <b>Controlled recursion</b>     |
| <b>RecursiveGraphNode</b> | <b>Recursive graph</b>          |
| <b>NestedGenericModel</b> | <b>Nested generics</b>          |
| <b>DeepGenericModel</b>   | <b>Deep generics</b>            |
| <b>MapEdgeCaseModel</b>   | <b>Map key/value edge cases</b> |

Test: GhostRobustnessTest, GhostAdvancedTypesTest, GhostTypeSystemTest

## C. JVM fragmentation (>40 fields)

| Model            | Approx. fields | Emitter                                                   |
|------------------|----------------|-----------------------------------------------------------|
| <b>----</b>      | <b>----</b>    | <b>----</b>                                               |
| <b>Object40</b>  | <b>40</b>      | <b>StandardEmitter (limit)</b>                            |
| <b>Object41</b>  | <b>41</b>      | <b>FragmentedEmitter</b>                                  |
| <b>GodObject</b> | <b>massive</b> | <b>FragmentedEmitter +<br/>FragmentedSerializeEmitter</b> |
| <b>WideModel</b> | <b>wide</b>    | <b>Serialize fragmentation</b>                            |

Test: GhostBoundaryFragmentationTest, GhostGodObjectTest

## D. Structural transformations

| Model                           | Key annotation          |
|---------------------------------|-------------------------|
| <b>----</b>                     | <b>----</b>             |
| <b>FlattenedModel</b>           | <b>@GhostFlatten</b>    |
| <b>DeepFlattenedModel</b>       | <b>deep flatten</b>     |
| <b>WrappedModel</b>             | <b>@GhostWrap</b>       |
| <b>MixedStructuralModel</b>     | <b>mix flatten/wrap</b> |
| <b>WrapSharedPathModel</b>      | <b>shared paths</b>     |
| <b>StructuralCollisionModel</b> | <b>path collisions</b>  |
| <b>CollisionChild</b>           | <b>collision child</b>  |
| <b>CollisionModel</b>           | <b>key collision</b>    |

Test: GhostStructuralTransformationTest

## E. Polymorphism with discriminator

| Root / subclasses | Discriminator |
|-------------------|---------------|
|-------------------|---------------|

|                                                                    |                       |
|--------------------------------------------------------------------|-----------------------|
| ---                                                                | ---                   |
| SmartDevice → Light, Thermostat, UnknownDevice                     | type + @GhostFallback |
| ApiEventDefault → Login, Logout                                    | implicit default      |
| ApiEventExplicitType → Login, Logout                               | explicit type         |
| GhostKindEvent → Created, Deleted, Updated                         | kind                  |
| StripeObject → Charge, Refund                                      | object                |
| JsonLdNode → Person, Organization                                  | JSON-LD @type         |
| GhostShape → Circle, Square                                        | geometric sealed      |
| GhostDiscriminatorTestPayload                                      | test payload          |
| Test: GhostCustomDiscriminatorTest, GhostResilienceAndFallbackTest |                       |

## F. Inferred polymorphism (test sources only)

| Model                                                         | Test                          |
|---------------------------------------------------------------|-------------------------------|
| ---                                                           | ---                           |
| SmartEvent, TempEvent, HumidityEvent, MixedEvent, MotionEvent | GhostInferredPolymorphismTest |
| DeviceCommand → Reboot, SetBrightness, UpdateFirmware         | GhostInferredPolymorphismTest |
| InferredNestedContainer                                       | nested inferred               |
| HugeModel, DeepNestedModel, MassiveInferredRoot               | GhostProductionHardeningTest  |

Generated under: build/generated/ksp/test/kotlin/

## G. Resilience, coercion, and enums

| Model                                | Behavior              |
|--------------------------------------|-----------------------|
| ---                                  | ---                   |
| ResilientItem                        | @GhostResilient items |
| DeepResilientModel                   | deep resilient        |
| SmartHome, HomeConfig, HomeStatus    | IoT resilient         |
| ResilientEnumModel                   | unknown enum          |
| CoercionStressModel                  | strings→numbers       |
| BooleanCoercionModel                 | 0/1 → bool            |
| GhostStandardsEnum, GhostEnumWrapper | enum standards        |

Test: GhostCoercionTest, GhostEnumResilienceTest, GhostEnumStandardsTest

## H. Custom coders and external types

| Model                                                  | Mechanism                          |
|--------------------------------------------------------|------------------------------------|
| ---                                                    | ---                                |
| CustomCoderStressModel                                 | @GhostEncoder/@GhostDecoder stress |
| CustomDateUser, LegacyUser                             | legacy dates                       |
| EncoderHexUtils, EncoderDateUtils, EncoderBooleanUtils | providers                          |
| ContextualModel, ModelWithExternal                     | 3rd-party types                    |
| ExternalColor + ExternalColorSerializer                | manual serializer                  |
| ExternalDate + ExternalDateSerializer                  | contextual date                    |

Test: GhostCustomCoderTest, GhostContextualTest

## I. Value classes and tokens

| Model                     | Detail                 |
|---------------------------|------------------------|
| ----                      | ----                   |
| <b>UserId</b>             | @JvmInline value class |
| <b>UserWithValueClass</b> | transparent wrapper    |
| <b>GhostUserToken</b>     | typed token            |

Test: GhostValueClassTest

## J. Nullability, defaults, evolution

| Model                         | Detail              |
|-------------------------------|---------------------|
| ----                          | ----                |
| <b>NullabilityStressModel</b> | null on all types   |
| <b>DefaultValueNullModel</b>  | default + null      |
| <b>NullablePrimitives</b>     | nullable primitives |
| <b>CollectionOfNulls</b>      | List with nulls     |
| <b>EvolutionModel</b>         | new/old fields      |
| <b>IgnoreModel</b>            | @GhostIgnore        |

Test: GhostNullabilityStressTest, GhostHardeningTest, GhostFinalHardeningTest

## K. Naming, Unicode, reserved words

| Model                    | Detail              |
|--------------------------|---------------------|
| ----                     | ----                |
| <b>NamingModel</b>       | @GhostName aliases  |
| <b>ReservedWordModel</b> | reserved JSON names |
| <b>UniCodeModel</b>      | complex UTF-8       |
| <b>EmojiKeyModel</b>     | emoji keys          |

Test: GhostFeatureTest, GhostEliteHardeningTest

## L. Malice, DoS, and large strings

| Model                                    | Detail             |
|------------------------------------------|--------------------|
| ----                                     | ----               |
| <b>MaliceModel</b>                       | malicious payloads |
| <b>LargeStringModel, LargeStringData</b> | huge strings       |
| <b>StressMetrics</b>                     | stress metrics     |
| <b>DecimalStress</b>                     | decimal precision  |

Test: GhostMaliceTest, GhostProductionHardeningTest

## M. Concurrency and multi-branch

| Topic                                | Detail                                             |
|--------------------------------------|----------------------------------------------------|
| ----                                 | ----                                               |
| <b>BenchUser</b>                     | 3 parameters with default → 8 constructor branches |
| <b>GhostConcurrencyExpansionTest</b> | pools + threads                                    |



Test: GhostMultiBranchConstructorTest, GhostConcurrencyExpansionTest

## Test class → responsibility map

| Test class                                          | Focus                         |
|-----------------------------------------------------|-------------------------------|
| ----                                                | ----                          |
| <b>GhostStressTest</b>                              | <b>throughput / stability</b> |
| <b>GhostRobustnessTest</b>                          | <b>rare types</b>             |
| <b>GhostBoundaryFragmentationTest</b>               | <b>40/41 fields</b>           |
| <b>GhostGodObjectTest</b>                           | <b>GodObject</b>              |
| <b>GhostStructuralTransformationTest</b>            | <b>flatten/wrap</b>           |
| <b>GhostInferredPolymorphismTest</b>                | <b>inferred</b>               |
| <b>GhostCustomDiscriminatorTest</b>                 | <b>custom discriminator</b>   |
| <b>GhostResilienceAndFallbackTest</b>               | <b>resilient + fallback</b>   |
| <b>GhostCoercionTest</b>                            | <b>type coercion</b>          |
| <b>GhostCustomCoderTest</b>                         | <b>encoder/decoder</b>        |
| <b>GhostContextualTest</b>                          | <b>external</b>               |
| <b>GhostValueClassTest</b>                          | <b>inline classes</b>         |
| <b>GhostNullabilityStressTest</b>                   | <b>nulls</b>                  |
| <b>GhostMaliceTest</b>                              | <b>parser security</b>        |
| <b>GhostProductionHardeningTest</b>                 | <b>massive inferred</b>       |
| <b>GhostMultiBranchConstructorTest</b>              | <b>bitmask branches</b>       |
| <b>GhostConcurrencyExpansionTest</b>                | <b>threads</b>                |
| <b>GhostLibraryMethodTest</b>                       | <b>API parity</b>             |
| <b>GhostTypeSystemTest</b>                          | <b>Map/List edge</b>          |
| <b>GhostAdvancedTypesTest</b>                       | <b>generics</b>               |
| <b>GhostFeatureTest</b>                             | <b>naming</b>                 |
| <b>GhostEliteHardeningTest</b>                      | <b>unicode</b>                |
| <b>GhostHardeningTest / GhostFinalHardeningTest</b> | <b>evolution</b>              |
| <b>GhostEnumResilienceTest /</b>                    | <b>enums</b>                  |
| <b>GhostEnumStandardsTest</b>                       |                               |

## VIII.2 Walkthrough: BenchUserSerializer line by line

### Source model (KSP input)

```
@GhostSerialization
data class BenchUser(
    val id: Int,
    val name: String,
    val email: String,
    val score: Double,
    val isActive: Boolean = true,
    val role: UserRole = UserRole.VIEWER,
    val bio: String? = null
)
```

Required JSON fields: id, name, email, score (bits 1+2+4+8 = mask

15L).

Optional with Kotlin default: `isActive`, `role`, `bio` → compiler generates multi-branch (8 constructor combinations).

### Generated output (location)

`build/generated/ksp/main/kotlin/com/ghost/serialization/integration/mode/l/BenchUserSerializer.kt`

### Block 1 — Header and contract (L1-38)

```
@file:Suppress("UNUSED_VARIABLE", "UNCHECKED_CAST", "UNUSED_EXPRESSION", ...)
@file:OptIn(InternalGhostApi::class)
```

```
public object BenchUserSerializer : GhostSerializer<BenchUser> {
    override val typeName: String = "BenchUser"
```

- object (not class): zero alloc on lookup; singleton.
- `typeName`: key in `GhostModuleRegistry` and `getSerializerByName`.

### Block 2 — `JsonReaderOptions` (L40-48)

```
private val OPTIONS = JsonReaderOptions.of(0, 31,
    "id", "name", "email", "score", "isActive", "role", "bio")
```

- 1024-slot trie; hash of first 4 bytes of field name.
- At runtime: `selectNameAndConsume(OPTIONS)` returns index 0..6, -1 end, -2 unknown.

### Block 3 — Pre-encoded headers (L50-62)

```
private val H_ID: ByteString = "\"id\":".encodeUtf8()
private val H_EMAIL: ByteString = "\"email\":".encodeUtf8()
private val H_NAME: ByteString = "\"name\":".encodeUtf8()
// (and H_SCORE, H_ISACTIVE, H_ROLE, H_BIO the same way)
```

- `Serialize` writes `"id"`: without re-encoding UTF-8 per request.
- `writeField(H_ID, value.id)` concatenates header + value.

### Block 4 — `Deserialize: variables and mask` (L67-76)

```
var _id: Int = 0
var _name: String? = null
var _role: UserRole? = null
var _bio: String? = null
var _mask0 = 0L
reader.beginObject()
```

- One temporary variable per property.
- `_mask0` accumulates bits: `id=1`, `name=2`, `email=4`, `score=8`, `isActive=16`, `role=32`, `bio=64`.

### Block 5 — Field loop (L77-114)

```
while (true) {
    val index = reader.selectNameAndConsume(OPTIONS)
```

```

when (index) {
    0 -> { _id = reader.nextInt(); _mask0 = _mask0 or 1L }
    1 -> { _name = reader.nextString(); _mask0 = _mask0 or 2L }
    5 -> { _role = UserRoleSerializer.deserialize(reader); _mask0 = _mask0 or 32L }
    6 -> { _bio = if (reader.isNextNullValue()) { reader.consumeNull(); null }
           else reader.nextString(); _mask0 = _mask0 or 64L }
    -1 -> break
    -2 -> reader.skipValue()
}
}
reader.endObject()

```

- Delegation: UserRole uses its own UserRoleSerializer (generated enum).
- Nullable: explicit isNextNullValue branch for bio.
- Unknown: skipValue() forward-compatible.

## Block 6 — Required field validation (L116-126)

```

if ((_mask0 and 15L) != 15L) {
    if ((_mask0 and 1L) == 0L) reader.throwError("Required field 'id' missing in JSON")
    else if ((_mask0 and 2L) == 0L) reader.throwError("Required field 'name' missing in JSON")
    // (same for email and score)
}

```

- One AND on aggregated mask; specific message for first missing field.
- Throws GhostJsonException with buffer position.

## Block 7 — Multi-branch constructors (L127-200)

```

if ((_mask0 and 112L) == 112L) {
    return BenchUser(id=_id, name=_name!!, email=_email!!, score=_score,
                     isActive=_isActive, role=_role!!, bio=_bio)
}
if ((_mask0 and 48L) == 48L) {
    return BenchUser(id=_id, name=_name!!, email=_email!!, score=_score, isActive=_isActive,
                     role=_role!!)
}
// Other branches: 112L (all optional), 80L, 96L, 16L, 32L, 64L...
return BenchUser(id=_id, name=_name!!, email=_email!!, score=_score) // required only

```

- 112L = 16+32+64 → isActive + role + bio present.
- Each branch calls the primary constructor once with the exact combination of arguments present in JSON.
- Avoids chained .copy() on hot path (1.1.16+ optimization).

## Block 8 — Flat deserialize (L206-340)

Duplicate logic for GhostJsonFlatReader — same structure, different reader (contiguous byte array). Ghost.deserialize(bytes) uses this path

via pool.

## Block 9 — Streaming and flat serialize (L342-370)

```
override fun serialize(writer: GhostJsonFlatWriter, value: BenchUser) {
    writer.beginObject()
    writer.writeField(H_ID, value.id)
    writer.writeField(H_NAME, value.name)
    writer.writeField(H_EMAIL, value.email)
    writer.writeField(H_SCORE, value.score)
    writer.writeField(H_ISACTIVE, value.isActive)
    writer.name(H_ROLE)
    UserRoleSerializer.serialize(writer, value.role)
    if (value.bio != null) {
        writer.writeField(H_BIO, value.bio)
    }
    writer.endObject()
}
```

- Nullable optional fields: omitted if null (bio).
- Fields with Kotlin default but present: always serialized if not null.

## Block 10 — warmUp() (L372-385)

```
override fun warmUp() {

    deserialize(GhostJsonReader("""{"id":0,"name":"","email":"","score":0.0}""").encodeToByteArray())

    deserialize(GhostJsonFlatReader("""{"id":0,"name":"","email":"","score":0.0}""").encodeToByteArray())
}
```

- Ghost.prewarm() invokes this to JIT/compile paths before real traffic.

## Full flow in your app

```
Ghost.deserialize<BenchUser>(json)
    → resolveSerializer → BenchUserSerializer (registry)
    → ghostInternalUseFlatReader(bytes) { reader ->
        BenchUserSerializer.deserialize(reader) // blocks 5-7
    }
```

## VIII.3 integration-test generated registry

File:

build/generated/ksp/main/kotlin/com/ghost/serialization/generated/GhostModuleRegistry\_Default.kt

```
// Conceptual structure (simplified)
class GhostModuleRegistry_Default : GhostRegistry {
    companion object { val INSTANCE = GhostModuleRegistry_Default() }
```

```
override fun <T : Any> getSerializer(clazz: KClass<T>): GhostSerializer<T>? =
    when (clazz) {
        BenchUser::class -> BenchUserSerializer as GhostSerializer<T>
        Address::class -> AddressSerializer as GhostSerializer<T>
        BenchUser::class -> BenchUserSerializer as GhostSerializer<T>
        Address::class -> AddressSerializer as GhostSerializer<T>
        else -> null
    }

    override fun getAllSerializers(): Map<KClass<*>, GhostSerializer<*>> by lazy {
        // full map for prewarm()
    }
}
```

ServiceLoader:

META-INF/services/com.ghost.serialization.contract.GhostRegistry points to this class.

Test registry: models in src/test generate  
GhostModuleRegistry\_Default\_Test (\_Test suffix when  
ghost.moduleName is Default and file is in test/).

Local inspection:

```
./gradlew :ghost-integration-test:compileKotlin
ls ghost-integration-test/build/generated/ksp/main/kotlin/com/ghost/serialization/
```

## 51. Maven artifacts table 1.2.2

```
com.ghostserializer:ghost-api
com.ghostserializer:ghost-serialization
com.ghostserializer:ghost-compiler (KSP)
com.ghostserializer:ghost-gradle-plugin
com.ghostserializer:ghost-retrofit
com.ghostserializer:ghost-ktor
com.ghostserializer:ghost-spring-boot-starter
```

Plugin id: com.ghostserializer.ghost version aligned with libraries.

## Factual verification (aligned with code 1.2.2)

This manual was cross-checked against the ghost-serializer repository on the local working branch:

| Manual claim                     | Source in repo                                   |
|----------------------------------|--------------------------------------------------|
| ---                              | ---                                              |
| Registry sharding at 500 entries | GhostEmitterConstants.REGISTRY_CHUNK_SIZE = 500  |
| Method fragmentation >40 fields  | DEFAULT_CHUNK_SIZE = 40 in GhostEmitterConstants |

|                                                   |                                                                                        |
|---------------------------------------------------|----------------------------------------------------------------------------------------|
| Multi-branch up to 4 defaults                     | MAX_DEFAULT_BRANCH_COUNT = 4 in StandardEmitter.kt                                     |
| maxCollectionSize Android 50k / JVM 1M / Nat 500k | GhostHeuristics.android.kt, .jvm.kt, .native.kt                                        |
| maxWarmWriteBuffer 4 MB mobile / 8 MB JVM         | same GhostHeuristics.*.kt files                                                        |
| maxDepth 255                                      | GhostJsonConstants.MAX_DEPTH                                                           |
| 416 tests in ciTestJvm (verified)                 | Gradle build/test-results XML, May 2026                                                |
| 642 tests ./gradlew ciTest on Linux               | 416 + 226 Android testDebugUnitTest                                                    |
| ~874 with iOS on macOS                            | README: 642 + iosSimulatorArm64Test (~232)                                             |
| Spring Boot 3.4.5 in tests                        | gradle/libs.versions.toml spring-boot = "3.4.5"                                        |
| Ktor 2.3.11                                       | gradle/libs.versions.toml ktor = "2.3.11"                                              |
| iOS without ServiceLoader                         | Ghost.ios.kt → discoverRegistries() = emptyList()                                      |
| JVM registry fast-path                            | Ghost.jvm.kt → Class.forName + ServiceLoader                                           |
| List/Map at runtime                               | Ghost.kt → ListSerializer, MapSerializer                                               |
| Set not at runtime                                | no SetSerializer in ghost-serialization (README may mention Set; verify release notes) |

If you upgrade the Ghost version, cross-check these files again before trusting exact numbers.

## Appendix: API Reference

This section documents the public API of Ghost Serialization, derived from the KDoc comments in the source code (version 1.2.2).

### A.1 Ghost — Main entry point

object Ghost in package com.ghost.serialization.  
Central entry point for Ghost Serialization. Manages modular registry discovery  
and serialization/deserialization across all platforms.

#### Serialization

| Method                 | Description                                                                                                                                                                      |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ---                    | ---                                                                                                                                                                              |
| serialize(sink, value) | Encodes value and writes the resulting JSON payload into the given Okio BufferedSink. Uses GhostJsonFlatWriter internally for a single bulk-write with no Okio segment overhead. |
| serialize(value)       | Convenience alias for encodeToString.                                                                                                                                            |
| encodeToString(value)  | Serializes value to an in-memory JSON String. Writes to a flat contiguous byte buffer and performs a zero-copy string decode at the end.                                         |

|                                         |                                                                                                                                           |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>encodeToBytes(value)</b>             | Serializes value to a UTF-8 JSON ByteArray. Skips intermediate string encoding/decoding.                                                  |
| <b>encodeAndDiscard(value)</b>          | Serializes value discarding the output. Useful for JIT/ART priming without needing the resulting bytes.                                   |
| <b>encodeToSink(sink, value)</b>        | Alias for <code>serialize(sink, value)</code> .                                                                                           |
| <b>encodeToSink(sink, value, clazz)</b> | Non-inline variant of <code>encodeToSink</code> for contexts where the type is only known as a KClass at runtime (e.g. Spring, Retrofit). |

## Deserialization

| Method                                       | Description                                                                                              |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------|
| <b>---</b>                                   | <b>---</b>                                                                                               |
| <b>deserialize&lt;T&gt;(json)</b>            | Deserializes a JSON String into an instance of type T.                                                   |
| <b>deserialize&lt;T&gt;(source)</b>          | Deserializes from an Okio BufferedSource stream. Reads all bytes eagerly into a reusable scratch buffer. |
| <b>deserialize&lt;T&gt;(bytes)</b>           | Deserializes a UTF-8 JSON ByteArray into an instance of type T.                                          |
| <b>deserialize&lt;T&gt;(json, options)</b>   | Advanced: Deserializes a JSON String with custom parser settings.                                        |
| <b>deserialize&lt;T&gt;(source, options)</b> | Advanced: Deserializes from a BufferedSource with custom parser settings.                                |
| <b>deserialize&lt;T&gt;(bytes, options)</b>  | Advanced: Deserializes a ByteArray with custom parser settings.                                          |
| <b>decodeFromBytes(bytes, clazz, limit)</b>  | Non-inline variant for reflection/framework contexts (Spring, Retrofit).                                 |
| <b>decodeFromSource(source, clazz)</b>       | Non-inline variant of <code>deserialize</code> for BufferedSource.                                       |

## Registry management

| Method                       | Description                                                                                                                                       |
|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>---</b>                   | <b>---</b>                                                                                                                                        |
| <b>addRegistry(registry)</b> | Registers a GhostRegistry manually. Critical on iOS, Wasm, and JS targets where ServiceLoader is unavailable.                                     |
| <b>getSerializer(clazz)</b>  | Resolves the GhostSerializer for a given class. Checks primitives first, then the fast-path cache, then registered modules.                       |
| <b>getSerializer(type)</b>   | Resolves the GhostSerializer for a KType, supporting generic types (List<T>, Map<K,V>).                                                           |
| <b>prewarm()</b>             | Triggers eager loading and JIT/ART warm-up cycles for all registered serializers. Call at app startup for zero-latency first-run deserialization. |
| <b>throwError(message)</b>   | Throws IllegalArgumentException. Utility for generated serializers.                                                                               |

## A.2 GhostRegistry — Module contract

interface GhostRegistry in package com.ghost.serialization.contract.

Registry interface for discovering and managing compiler-generated and custom serializers.

Implementations are typically generated automatically as module-level registries by the Ghost

compiler plugin, allowing reflection-free serializer lookup across all targets in a Kotlin

Multiplatform project.

| Method                      | Description                                                                                            |
|-----------------------------|--------------------------------------------------------------------------------------------------------|
| ---                         | ---                                                                                                    |
| <b>getSerializer(clazz)</b> | <b>Resolves a GhostSerializer for the given class. Returns null if not registered in this module.</b>  |
| <b>getAllSerializers()</b>  | <b>Returns all serializers registered in this module. Used by Ghost for eager loading and prewarm.</b> |
| <b>prewarm()</b>            | <b>Eagerly initializes registry entries. No-op by default; can be overridden.</b>                      |
| <b>registeredCount()</b>    | <b>Returns the total number of serializers in this registry.</b>                                       |

## A.3 GhostJsonException — Parsing exception

class GhostJsonException in package com.ghost.serialization.exception.

Exception type thrown for JSON parsing/encoding errors.

To keep the failure path cheap (the parser may raise this exception in tight loops

while probing payloads), line and column are computed lazily — the O(N) scan over the source bytes is only paid if the caller actually reads either property

or accesses message.

| Property / Constructor                          | Description                                                                                      |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------|
| ---                                             | ---                                                                                              |
| <b>path: String</b>                             | <b>The dot-separated JSON path where the error occurred. Defaults to "\$" (root).</b>            |
| <b>line: Int</b>                                | <b>The 1-indexed line number in the JSON source where the error occurred. Computed lazily.</b>   |
| <b>column: Int</b>                              | <b>The 1-indexed column number in the JSON source where the error occurred. Computed lazily.</b> |
| <b>message</b>                                  | <b>Full message: "&lt;msg&gt; [at line X, col Y, path Z]".</b>                                   |
| <b>constructor(message, line, column, path)</b> | <b>Secondary constructor with an explicit error location.</b>                                    |



## A.4 JsonReaderOptions — Optimized field dispatch

class `JsonReaderOptions` in package `com.ghost.serialization.parser`.

Dispatch options for optimized JSON field identification. Uses a 4-byte hashing

engine to minimize collisions during field lookup.

`rawBytes` stores field names as raw `ByteArray` instead of Okio's `ByteString`,

so that `verifyKeyMatch` can compare bytes directly without virtual dispatch

or redundant bounds checks inside Okio's `rangeEquals`.

| Factory                                                            | Description                                                                                                               |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>----</code>                                                  | <code>----</code>                                                                                                         |
| <code>JsonReaderOptions.of(vararg names)</code>                    | Creates an optimized configuration with default hashing parameters ( <code>shift=0</code> , <code>multiplier=31</code> ). |
| <code>JsonReaderOptions.of(shift, multiplier, vararg names)</code> | Creates a configuration with custom hashing shift and multiplier values to minimize collisions for a specific field set.  |

## A.5 @InternalGhostApi — Internal use annotation

annotation class `InternalGhostApi` in package `com.ghost.serialization`.

Marks declarations that are internal to Ghost Serialization.

Declarations annotated with this annotation are not intended for public use and may

change or be removed in future versions without notice. Code generated by the Ghost KSP

compiler plugin uses this annotation to access internal optimized helper functions.

Opt-in level: `RequiresOptIn.Level.WARNING`. User code that directly uses this API will receive a compiler warning.